

Modulation of excessive neuronal activity by fibroblasts: Potential use in treatment of Parkinson's disease

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Abstract. *Purpose:* A number of neurological disorders are marked by increased or aberrant frequency of neuronal discharge in specific parts of the brain. Administration of drugs such as antiepileptic compounds results in the depression of neuronal activity in the whole brain, with the potential for serious side-effects. In the search for additional therapies to reduce the unphysiological electrical activity of over-active brain foci, we have examined the effect of fibroblasts transplanted to areas responsible for motor dysfunction in hemi-parkinsonian rats, since bursting synchronous discharges in internal segment of globus pallidus (GPi) are thought to be partially responsible for the movement disorders of PD. Fibroblasts express gap junctions and ion channels, and so, when transplanted to brain tissue, can potentially modulate excessive electrical activity.

Methods: Neonatal cortical neurons were cultured on multi-electrode arrays, and their electrical activity was evaluated before and after fibroblast seeding. Unilateral 6-hydroxydopamine (6-OHDA) lesion was carried out in Fischer rats. Lesioned or control rats were transplanted with either syngeneic dermal fibroblasts, microfine glass beads, ibotenic acid, or physiological saline, in the entopeduncular nucleus (EP). Apomorphine-induced rotational behavior and L-dopa-induced dyskinetic movements were evaluated before transplantation (baseline) and 2, 4, 8, 12, and 24 weeks following transplantation. Following behavioral experiments, rats were perfused with 4% formaldehyde in PBS for immunohistochemical study of the brain.

Results: We demonstrate *in vitro* that the introduction of fibroblasts into a network of neurons does not interfere with overall functional measures of activity, while moderately altering the characteristics of synchronous neuronal discharge. In rats with unilateral 6-hydroxydopamine lesions of the nigro-striatal dopaminergic pathway, apomorphine-induced rotations were reduced by more than 60% following ipsilateral transplantation of fibroblasts to the EP. L-Dopa-induced dyskinesia was also significantly reduced. Transplantation of inert microspheres, or chemical lesion of the same area with ibotenic acid, did not produce beneficial effects on parkinsonian symptomatology.

Conclusion: Fibroblast transplantation could be an alternative treatment strategy for the parkinsonian patient.

Keywords: 6-hydroxydopamine, entopeduncular nucleus, L-dopa-induced dyskinesia, apomorphine-induced rotation, multi-electrode array, cell transplantation

1. Introduction

In previous studies we have demonstrated the efficacy of fibroblasts in modulating synchrony of cardiac networks *in vitro*, (Feld et al., 2002) and in treating cardiac arrhythmias *in vivo* (Yankelson et al., 2008). Since

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the addition of fibroblasts to cultures of cardiomyocytes was effective in preventing spread of abnormal electrical discharges, we decided to adapt the technique to neuronal systems. In previous studies, fibroblasts have been transplanted to human brain as vehicles for production of deficient molecules, such as neurotrophic factors (Gage et al., 1987; Levivier et al., 1995; Tuszynski et al., 2005) or L-dopa (Wolff et al., 1989), proving the safety of fibroblast transplantation to human brain. Parkinson's disease is an obvious target for cell therapy, since the disease symptoms are mainly the result of loss of a single type of neuron, namely dopamine-producing neurons of the nigro-striatal pathway, however previous attempts at cell replacement therapy of Parkinson's disease using dopaminergic cells have been unsuccessful for various reasons, including the development of severe dyskinesias (Lindvall and Hagell, 2002).

Measurements in brains of experimental animals after dopaminergic lesion, as well as in brains of human parkinsonian patients undergoing stereotactic surgery for implantation of stimulating electrodes or pallidotomy, indicate that the machinery underlying behavioral parkinsonian symptoms involves an increased tendency for synchronized bursting of neuronal discharges (Bergman et al., 1994; Plenz and Kital, 1999; Soares et al., 2004; Wichmann and DeLong, 2006). These bursts of synchronous discharges, which have been seen in sub-thalamic nucleus (STN), internal segment of globus pallidus (GPi) and pars reticulata of substantia nigra (SNpr) *in vivo*, (Wichmann and Soares, 2006) have also been demonstrated in primary neuronal cultures of STN and external segment of globus pallidus (Plenz and Kital, 1999), are a normal feature of primary cortical neuronal cultures, and are suppressed in the latter by direct electrical stimulation (Wagenaar et al., 2006). Evidence from human studies shows that electrical stimulation (DBS) of STN or GPi reduces synchronous bursting activity in the basal ganglia target neurons, by an as-yet unknown mechanism, (Brown et al., 2004; Meissner et al., 2005; Wingeier et al., 2006) and reverses parkinsonian symptoms.

Here we show the efficacy of fibroblasts in modifying the process of synchrony generation in neuronal networks *in vitro*, and in reduction of parkinsonian symptoms in a rat model of parkinsonism, after transplantation into the entopeduncular nucleus (EP, comparable to human GPi).

2. Methods

All experimental protocols in animals were approved by the Animal Care and Use Committee of the Tech-

nion. Fischer strain rats were used in all experiments except *in vitro* electrophysiology, in which both primary neurons and fibroblasts were prepared from Sprague-Dawley strain rats.

2.1. Preparation of dissociated dermal fibroblasts

Rat fibroblasts were isolated as described previously (Wang et al., 2004). Briefly tail skin was minced to small pieces (2–3 mm²), then digested in phosphate-buffered saline (PBS) containing 10 mg/ml collagenase type 1 (Worthington) for 3 h at 37°C with gentle shaking. After incubation, the digest was filtered through a 100 µm cell strainer, and rinsed with 10 ml growth medium (DMEM, Gibco) supplemented with 10% bovine calf serum (HyClone), 2.5 µg/ml amphotericin, 25 µg/ml gentamicin, and 2 mM glutamine (Biological Industries Ltd., Israel). The cells were seeded in 60 mm plates and grown to confluence, while the medium was changed every 3 d. The cells were expanded by passaging 1:3 until required cell quantities were obtained and then detached with 5% trypsin (Beit Haemek, Israel).

2.2. Histology

Rats were anesthetized deeply with pentobarbital and perfused via the left ventricle with 0.1 M phosphate-buffered saline (PBS, pH 7.4) followed by 4% paraformaldehyde in PBS. Brains were removed and fixation continued in the same paraformaldehyde/PBS solution at 4°C for 3 days. Coronal slices of the formalin-fixed brains were dehydrated in ethanol of increasing concentration followed by delipidation with xylene, and embedding in paraffin. For immunohistochemistry, deparaffinized and rehydrated slides were exposed to microwave radiation in 0.01M citrate buffer (pH 6.8), permeabilized in 0.5% Triton (X-100) in 0.02M phosphate buffer, treated with 0.3% hydrogen peroxide in absolute methanol to quench endogenous peroxidase, incubated in 10% goat non-immune serum blocking solution (Zymed, Invitrogen Corporation, USA), and then incubated with the following primary antibodies: mouse monoclonal anti-tyrosine hydroxylase antibody (Chemicon International, AB152), mouse anti-HSP 47 (Abcam, ab13510), mouse anti-Iba1 (Abcam, ab 15690), rabbit anti-CD3 (Abcam, ab5690), or monoclonal anti-glial fibrillary acidic protein (Sigma) in PBS with 0.1% Triton X-100, 5% bovine serum albumin and 5% fetal calf serum for 18 h at 4°C. Primary antibodies were visualized with appropriate anti-mouse or

anti-rabbit HRP-conjugated secondary antibodies and 3-amino-9-ethylcarbazole (AEC) chromogen reaction, and counterstained with hematoxylin (Zymed, Invitrogen Corporation, USA).

For measurement of lesion size, 5 μm coronal sections were made each 100 μm from levels -2.0 to -3.4 mm posterior to bregma and stained with hematoxylin and eosin. Morphometric analysis was performed using the Cavalieri approach (Mouton, 2002). Measurement of lesion areas was performed using an eyepiece micrometer. The length (μm) of the longest axis of implant was measured initially, then a second measurement was made at 90° to the first. The volume of the lesion was calculated as the sum of elliptical cylinders, where for each section, cylinder volume $V = n(\pi a b) \times 100 \mu\text{m}^3$, where n = number of slides, a = length of long axis/2, b = length of short axis/2.

2.3. Electrophysiology of cultured cortical neurons

Primary cultures of rat cortical neurons were prepared as described previously (Eytan et al., 2003; Eytan and Marom, 2006; Marom and Shahaf, 2002; Shahaf and Marom, 2001). Briefly, cortical neurons were obtained from 1 d postnatal Sprague Dawley rats, (Harlan, Israel). The cortical tissue was enzymatically digested, mechanically dissociated, and the neurons were plated directly onto substrate-integrated multi-electrode array (MEA) dishes (Gross, 1979). Cultures were grown in MEM (Biological Industries Ltd., Israel) supplemented with heat-inactivated horse-serum (5%), glutamine (0.5 mM), glucose (20 mM), and gentamycin (10 $\mu\text{g}/\text{ml}$), and maintained in a tissue culture incubator in an atmosphere of 5 % CO_2 and 95 % air at 37°C . There were about $1 \cdot 10^4$ cells in a mature cortical neuron culture. Once the cultures expressed stable neuronal discharge and following baseline recording, $5 \cdot 10^4$ fibroblasts were added to the culture. At the third day after fibroblasts seeding the array was placed on the recording system and the electrical activity of the co-culture was recorded using the same parameters as the base-line recordings. During the recording phases the cultures were exposed to the same environmental conditions using a miniature incubation cell built onto the microscope stage. Electrophysiological measurements were made during the third week in vitro, allowing for a complete structural and functional network maturation, and synchronous events were detected as described before (Eytan and Marom, 2006) using an activity threshold. A timestamp is defined when activity crosses a threshold of N number of action potentials

recorded throughout the electrode array within a T ms time bin; N is equal to 1/4 of the (active) electrodes, and $T = 3$ ms. An active electrode is defined as demonstrating an average firing rate of $> 0.02 \text{ s}^{-1}$ throughout the recording session. The data around each timestamp (± 300 ms) is extracted and stored. Temporal overlap between synchronous events was not allowed.

2.4. 6-OHDA lesioning

6-OHDA lesion was carried out using the general method described by others (Henry et al., 1999; Lundblad et al., 2002; Ungerstedt, 1971). Briefly, desipramine HCl (10 mg/kg, s.c.) was administered 30 min prior to the stereotaxic injection of 8 $\mu\text{g}/4\mu\text{l}$ 6-OHDA (in physiological saline solution containing 0.1% ascorbic acid) into the right medial forebrain bundle of Fischer rats. Injection coordinates, relative to bregma, were: AP -4.4 ; LR 1.3; DV -8.1 (from skull). The 6-OHDA solution was infused at $1\mu\text{l}/\text{min}$ for 4 min. At the end of the injection the needle (30G with rounded tip) was left *in situ* for an additional 5 min and then slowly retracted.

2.5. Syngeneic Fischer dermal fibroblast transplantation

Fischer strain rats were injected with 100,000 Fischer dermal fibroblasts in PBS supplemented with 1mg/ml glucose to the entopeduncular nucleus (EP, corresponds to GPi in human). Coordinates used for EP (relative to bregma and skull) were: AP-2.8, LR +2.9, DV -7.7). Cells were injected over a period of 4 min at a rate of $1 \mu\text{l}/\text{min}$, and the needle was then slowly removed as described above. Control animals were infused with the same volume of PBS-glucose. In other control experiments, inert polystyrene microspheres (Fluospheres, 10 μm diameter, Invitrogen), were implanted to the EP (100×10^3 microspheres per $4 \mu\text{l}$ PBS-glucose infused over a period of 4 min), or the neurotoxin ibotenic acid was injected locally (24 nmol in $0.2 \mu\text{l}$ injected over 5 min as described by Inglis and Semba, 1997).

2.6. Rotation test

Four weeks after 6-OHDA lesion each animal was screened for apomorphine-induced rotational asymmetry. Rats were placed in a cylindrical cage and injected with 0.25 mg/kg s.c. apomorphine (Sigma) after a 15 min acclimatization period. Numbers of rotations were recorded at 5 min intervals using an automatic ro-

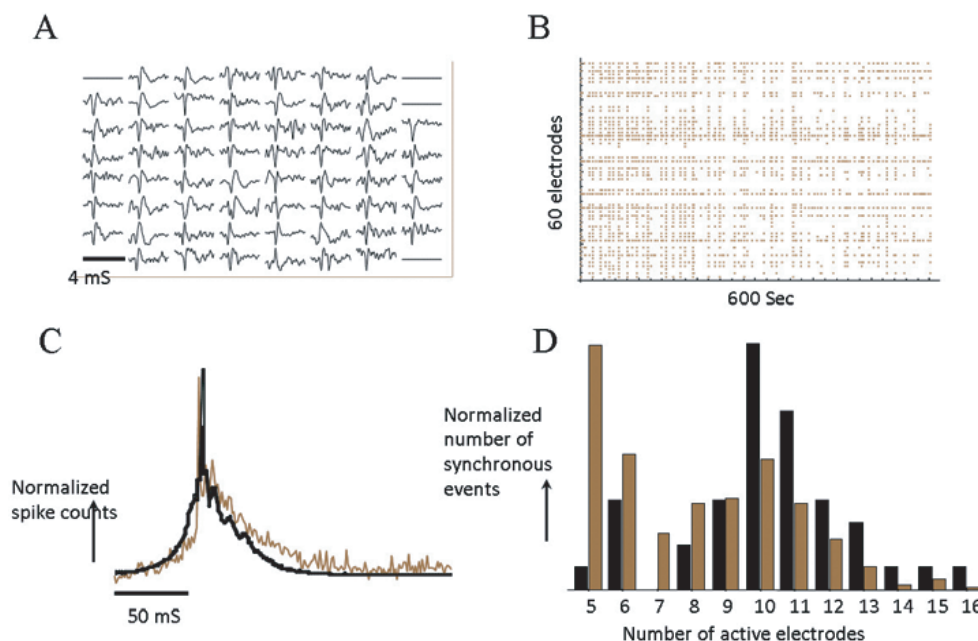


Fig. 1. Synchronization of the electrical discharge of cultured rat cortical neurons: (A) example of multiple electrode recording during a single synchronous event; (B) spikes in all 60 electrodes in multiple array dish over 10 min; vertical lines show synchronies; (C) summated spikes recorded from cultured cortical neurons before (black line) and 3 days following seeding of dermal fibroblasts (grey line); (D) normalized histograms of the number of electrodes that detected spikes at the peak of a synchronous event. Black – no fibroblasts. Brown – with fibroblasts. Note the left shift and clear bi-modal shape of the histogram in the presence of fibroblasts, indicating an increase in the fraction of low amplitude synchronization events. The histograms summarize stationary time series that include more than 600,000 spikes and over 400 synchronous events.

tometer (Rota-Count-8; Columbus Instruments) over a 1.5 h period. Rats were accepted in the test after meeting the threshold criterion of reaching a maximum value of 35 full contralateral rotations in any 5 min period within the 1.5 h after injection. Apomorphine-induced contralateral rotations were similarly determined over 1.5 h in response to the same dose of apomorphine administered at the time periods shown after cell or vehicle transplantation. Average rotational behavior was expressed as mean rotations per 5 min, and peak rotations were similarly expressed as mean rotations per 5 min for three 5 min periods at the peak of rotational activity after apomorphine. On the basis of our own experience, and published work from other laboratories (Schwartz and Huston, 1996; Spooren et al., 1999), animals satisfying the above criterion will have 95% or greater loss of dopaminergic neurons in the substantia nigra, and greater than 95% depletion of striatal dopamine content on the lesioned side.

2.7. Dyskinesia test

Dyskinesia induced by L-dopa was evaluated as described previously by others (Lundblad et al., 2002).

In one series of experiments (results of apomorphine-induced turning shown in Fig. 2b), L-dopa-induced dyskinesias were examined following completion of the 24 weeks follow-up after fibroblast implantation to EP. Four weeks after the last of the apomorphine-induced rotation tests shown in Fig. 2b, daily injections of L-dopa methyl ester (6 mg/kg ip) and carbidopa (5 mg/kg ip) were commenced for 1 week. Dyskinetic movements were scored on days 1, 4 and 7. In an additional series of experiments shown in Fig. 2d, in which the effects of fibroblasts, microspheres or ibotenic acid lesion were examined over a period of 2 months, treatment with L-dopa/carbidopa was started 2–3 months after the last apomorphine injection.

For dyskinesia evaluation rats were placed in a perspex cylinder (40 cm diameter) with cardboard base and observed every 20 min following CD/LD for the following behaviors:

1. axial abnormal involuntary movements (AIMs), ie dystonic or choreoform twisting of the neck and upper body towards the side contralateral to the lesion;
2. limb AIMs, ie abnormal or purposeless movements of the forelimb and digits contralateral to the lesion;

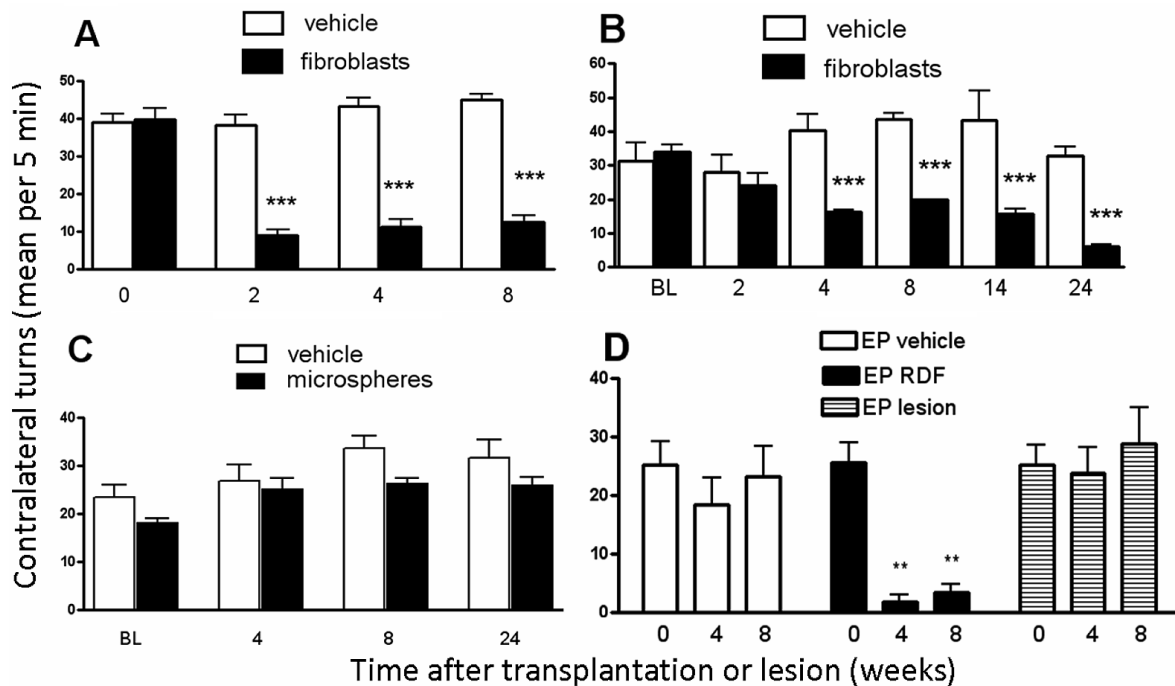


Fig. 2. Apomorphine-induced rotational behavior in rats with unilateral 6-OHDA lesion of nigro-striatal bundle: (A) following ipsilateral transplantation of syngeneic dermal fibroblasts (filled columns) or vehicle (open columns) to EP and SNr ($n = 6$, fibroblasts; $n = 4$, vehicle); (B) following ipsilateral transplantation of syngeneic dermal fibroblasts (filled columns) or vehicle (open columns) to EP ($n = 11$, fibroblasts; $n = 4$, vehicle); (C) following ipsilateral transplantation of microspheres (closed columns) or vehicle (open columns) to EP ($n = 7$, microspheres; $n = 4$, vehicle); (D) following ipsilateral transplantation of fibroblasts (closed columns), vehicle (open columns) or ibotenic acid (shaded columns) to the EP ($n = 9$ rats in each group). Data shown are means $\pm$ s.e.m. of complete contralateral turns per 5 min over total period of observation of 85 min. Time after cell, microsphere transplantation or ibotenic lesion shown on x-axis in weeks. Significance of differences calculated using one-way ANOVA followed by Bonferroni's multiple comparison test. ** $P < 0.01$, *** $P < 0.001$.

3. orolingual AIMs, ie empty jaw movements and contralateral tongue protrusion;
4. locomotor AIMs, ie increased locomotion with contralateral side bias;
5. abnormal head movements;

Each movement type is scored as follows:

- 1 = weak movements, present less than half of observation period;
- 2 = moderate movements present half of observation period;
- 3 = strong movements present more than half of observation period and suppressed by hand clap;
- 4 = strong movements present more than half of observation period not suppressed by hand clap.

Assessment of dyskinesia was performed by a single trained observer who was blinded to the treatment of the animal being observed.

2.8. Statistical analysis

Results are expressed as mean $\pm$ s.e.m. for the rotational and dyskinesia studies and mean $\pm$ s.d. for the electrophysiological parameters. The null hypothesis was rejected at a probability value less than 5%. Rotational behavior and dyskinesia analysis data was analysed using one-way ANOVA with Bonferroni test post hoc. Means of electrophysiological data were compared using Student's t-test.

3. Results

3.1. Modification of the electrophysiological properties of cortical neuronal network by fibroblasts

We initially examined the effect of fibroblast addition to neuronal cultures using a network of primary cortical neurons. These networks generate spontaneous and

stimulus-evoked synchronous events, a feature that is highly relevant in the context of the present study, and is well characterized (Eytan and Marom, 2006; Marom and Shahaf, 2002; Wagenaar et al., 2006), providing a baseline for comparison.

Figure 1A shows a typical 60 channels recording panel during a single synchronous event, where practically all the electrodes sense electrical discharges within the same time window. Figure 1B shows a low-resolution representation of network activity over 10 min of recording in the presence of fibroblasts, demonstrating the strong tendency of these networks to synchronize. Our analyses show that the introduction of fibroblasts into living neuronal cultures did not result in an appreciable change in overall time-amplitude envelope of neuronal synchronous events (Fig. 1C). However a closer examination suggests that several of the fine details of the activity are affected in a manner that might be reflected in a systemic, whole-brain level: in five different networks that were exposed to 17 recording episodes, each of which lasted 30–300 min, the mean rate of synchrony appearance was 0.048 s^{-1} (s.d. = 0.03), slightly lower ($p = 0.28$) compared to the rate of synchronous bursts in the absence of fibroblasts (0.08 s^{-1} , s.d. = 0.06; $n = 13$; see also Eytan and Marom, 2006). More significantly, the rates at which neurons are recruited within each synchronous event, extracted by fitting an exponential function to the initial recruitment phase (Eytan and Marom, 2006), showed an approximately 3 fold increase ($p = 0.0003$) in the presence of fibroblasts compared to the rate in the absence of fibroblasts. Specifically, in the presence of fibroblasts, during the build-up of synchrony, each millisecond the population of synchronized neurons increases by 12% compared to 4.5% in their absence. Given the exponential nature of the recruitment process during synchrony, the above 3-fold difference translates to speeding-up of synchrony timescale by ~ 50 milliseconds, a significant effect indeed. In addition, analysis of individual synchronous events showed that the presence of fibroblasts caused an alteration of the spectral distribution of the numbers of neurons per event, with an increase in the numbers of synchronies with a small number of neurons (Fig. 1D).

3.2. Transplantation of fibroblasts to basal ganglia of 6-OHDA-lesioned rats reduces rotational activity following apomorphine injection

In preliminary experiments, we examined the effect of fibroblast transplantation to STN, to both EP and

SNpr, and only to EP. Significant and consistent reductions in apomorphine-induced turning were obtained following transplantation to EP alone, whereas transplantation to STN produced a smaller and less significant degree of turning reduction. Combined transplantation to EP and SNpr also caused highly significant reductions in turning. Since transplantation to EP alone was effective, subsequent experiments were carried out using this site alone.

The effect of fibroblast transplantation to EP and SNpr on apomorphine-induced turning is shown in Fig. 2A. Apomorphine was administered 2, 4 and 8 weeks after fibroblast transplantation. Control rats showed a consistent level of apomorphine-induced turning over the 8 week period, while in fibroblast-transplanted rats, mean turning rate was reduced by $73 \pm 9.4\%$ at 2 weeks after transplantation, and maintained at the same low level throughout the 8 week period ($P < 0.001$ for difference from control at each time interval). In 4 of the 6 transplanted rats, the same low level of contralateral rotation was seen 6 months after transplantation, while the 2 surviving control rats also showed the expected higher rate of turning (not shown in Fig. 2A). Peak turning rates were reduced to the same extent as mean turning rate following fibroblast transplantation (70.67 ± 8.9 before, 21.3 ± 7.0 , 26.1 ± 8.6 , 24.3 ± 5.7 turns/5 min at 2, 4 and 8 weeks after transplantation; means $\pm$ s.e.m.).

In a subsequent experiment (Fig. 2B), we examined the effect of fibroblast transplantation to EP only. Eleven Fischer rats were injected with 100,000 Fischer dermal fibroblasts to the EP and vehicle was injected to the EP of 7 rats. The EP study group demonstrated more than 60% reduction in the rotational behavior compared to vehicle during the 6 months follow-up period ($P < 0.001$ at each time point for difference between fibroblasts and vehicle).

To exclude the possibility that the reductions in turning were the result of physical obstruction of electrical conduction in the appropriate basal ganglia, synthetic microspheres, of a size similar to that of granular neuron cell bodies (mean diameter $10 \mu\text{m}$), were injected to the EP of 7 rats with unilateral 6-OHDA lesions. No statistically significant change in rotational behavior was observed over a follow-up period of 6 months (Fig. 2C). In an additional control experiment, a discrete lesion of the EP was carried out by injection of ibotenic acid. No alteration in turning was seen up to 2 months after lesioning (Fig. 2D) while fibroblast transplantation caused a marked reduction in turning rate as before (data in Fig. 2D was obtained following transplantation of the same number of RDF but with an improved injection technique).

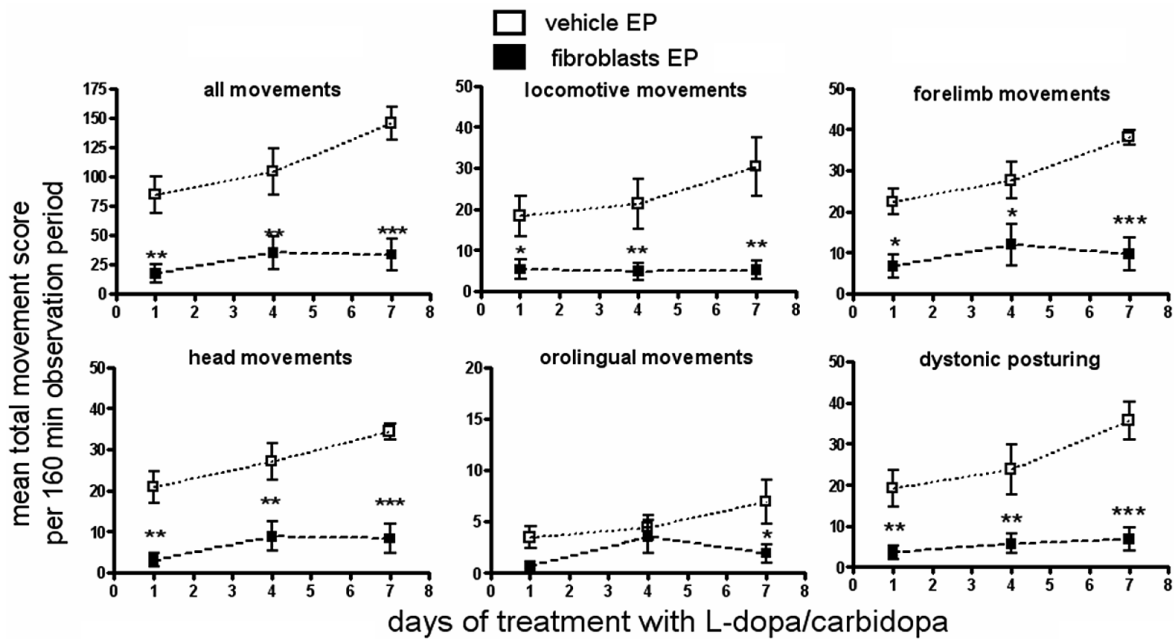


Fig. 3. L-Dopa-induced dyskinetic movements in rats with unilateral nigro-striatal 6-OHDA lesion, 7 months following subsequent ipsilateral transplantation of syngeneic dermal fibroblasts into the EP. The peripheral decarboxylase inhibitor carbidopa (5 mg/kg ip) was injected together with L-Dopa methyl ester (6 mg/kg ip) daily and dyskinetic movements evaluated over a period of 160 min following injection on days 1, 4 and 7. Data shown are means $\pm$ s.e.m. for $n = 11$ rats (fibroblasts) or $n = 6$ (vehicle). Significance of differences between fibroblasts and vehicle groups was determined using one-way ANOVA with Bonferroni post test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

3.3. Dyskinesia

At 7 months post 6-OHDA lesion and subsequent cell transplantation into the EP (Fig. 3), and upon completion of 6 months follow-up rotation evaluation, the animals received chronic treatment with carbidopa and L-Dopa, commencing 4 weeks after the last apomorphine test, and recordings of abnormal involuntary movements (AIMs) were carried out on days 1, 4 and 7 during this week. The presence of fibroblasts in the EP decreased all L-Dopa-induced dyskinetic movements significantly, with a smaller effect on orolingual movements. Following fibroblast transplantation to STN, dyskinetic movements were not significantly affected (data not shown). In control experiments with microsphere injection or ibotenic acid lesion, dyskinetic movements were also not significantly altered (Fig. 4). These results indicate that fibroblast implant to EP normalized the asymmetry of the hemiparkinsonian rat, as well as markedly reducing L-Dopa-induced dyskinetic movements.

4. Histology

In a separate toxicological study, rats were sacrificed 3 and 9 months after cell transplantation for full

pathology by a qualified neuropathologist. No tumors or pathological changes were observed throughout the sectioning of the whole brain (data not shown). In the cell-transplanted specimens the injected fibroblasts were identified in the proper location and distinguished from the surrounding brain tissue by unique fibroblast spindle-like shape (Fig. 5A). The approximate size of lesions produced in EP following fibroblast and microsphere implantation, as well as ibotenic acid lesion, were determined by stereology. The volume of tissue affected in the three types of lesion (see Fig. 5) were not significantly different: fibroblasts 12.4 ± 1.9 , microspheres 12.9 ± 3.5 , ibotenic acid 11.4 ± 3.1 ($\mu\text{m}^3 \times 10^6$, mean $\pm$ SEM).

The survival and viability of the transplanted fibroblasts were confirmed using immuno-histochemistry for the specific fibroblast marker HSP 47 (Fig. 6A). The area of fibroblast transplantation was surrounded by a narrow rim of GFAP-positive cells (Fig. 6C). A similar picture of gliosis was seen surrounding the area of ibotenic acid lesion, but vehicle-injected samples showed no evidence of astrocyte recruitment. Slices were also taken for immunohistochemical analysis of inflammation markers. No active inflammation was observed in fibroblast transplanted areas, as shown by only minor staining for the inflammation markers CD3 (for

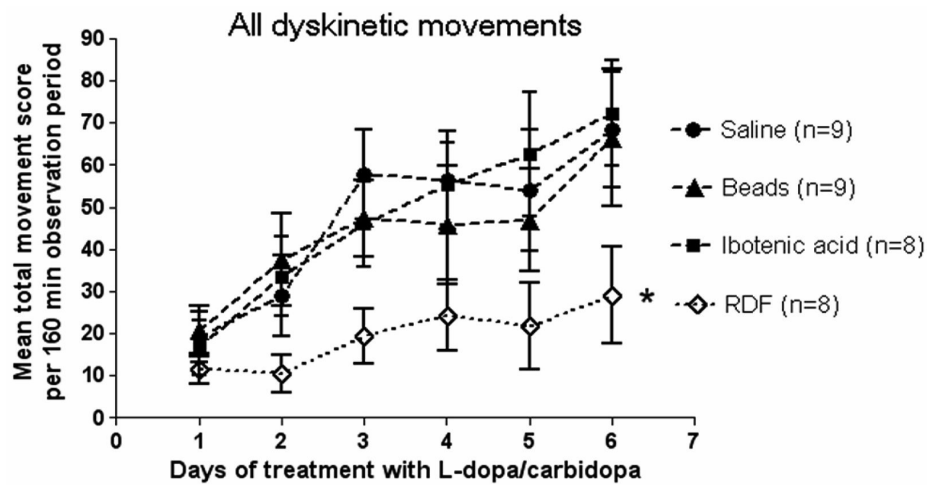


Fig. 4. L-Dopa-induced dyskinetic movements in rats with unilateral nigro-striatal 6-OHDA lesion, 7 months following subsequent ipsilateral transplantation of syngeneic dermal fibroblasts or microspheres, or ibotenic acid lesion, to the EP. The peripheral decarboxylase inhibitor carbidopa (5 mg/kg ip) was injected together with L-Dopa methyl ester (6 mg/kg ip) daily and dyskinetic movements evaluated over a period of 160 min following injection. Numbers of animals shown in parentheses. Only fibroblasts caused significant differences from vehicle using one-way ANOVA, * $P < 0.05$.

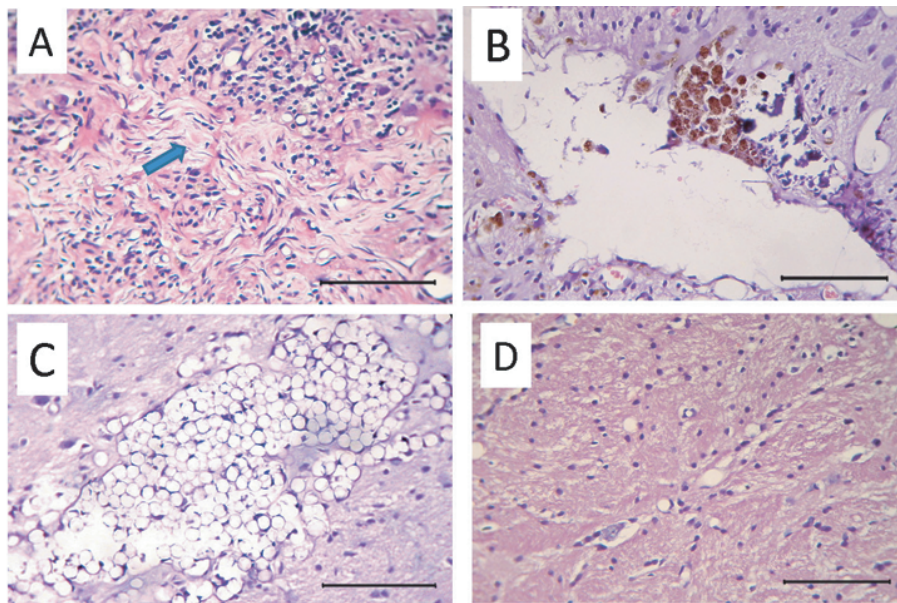


Fig. 5. Histological examination of EP, 6 months after: (A) rat dermal fibroblasts (RDF) implantation (fibroblasts can be distinguished by their characteristic elongated shape, shown by arrow); (B) ibotenic acid lesion; (C) microspheres implantation; (D) vehicle injection. Hematoxylin and eosin staining, 5 micron paraffin sections. Bar represents 100 microns.

T-lymphocytes, Fig. 6B), CD79a (for B-lymphocytes, not shown), and Iba1 (for activated microglia, Fig. 6D).

In order to confirm the persistence of stable unilateral striatal lesion throughout the entire experiment, anti-tyrosine hydroxylase (TH) staining of the SN was carried out. In the 6-OHDA-lesioned rats 3 and 9 months after the lesion and fibroblast transplantation there was

a marked loss of TH-immunoreactive neurons in the lesioned SN compared with the unlesioned side, commensurate with the $> 95\%$ loss of dopaminergic neurons seen in similar experiments. There was no difference in the appearance of SN between fibroblast- and vehicle-treated brains (see Fig. 7). Additionally, striatal dopamine content from both brain sides was measured

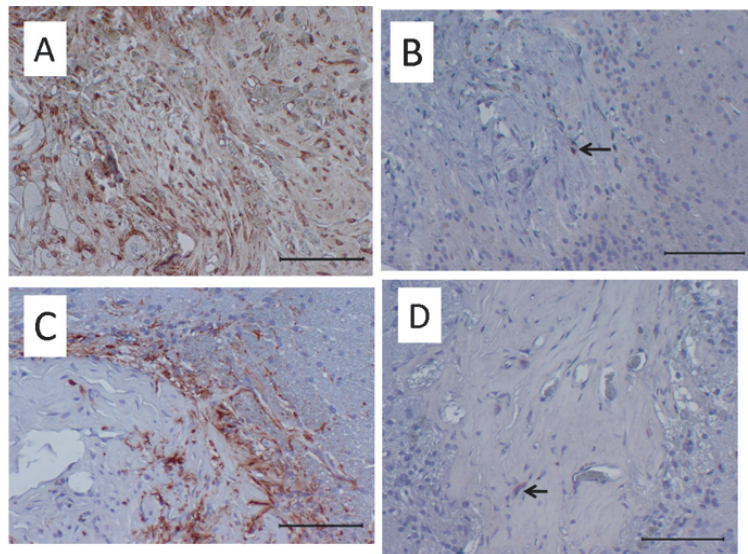


Fig. 6. Histological examination of EP, 6 months after implantation of rat dermal fibroblasts. Immunohistochemical staining for presence of: (A) HSP47, most fibroblasts show positive staining, as well as some cells outside the implant area; (B) CD3, very few cells are stained (black arrow); (C) GFAP, a band of astrocytes staining for GFAP can be seen bordering the implantation site; (D) Iba1, occasional cells positive for Iba1 are seen, indicated by black arrow. Horizontal calibration bar = 100 microns.

by HPLC analysis on 5 animals: 3 injected with vehicle and 2 with fibroblasts. In all animals the results demonstrated ~150–500 fold decrease in dopamine content in the lesioned compared with the unlesioned striatum.

5. Discussion

Using isolated neuronal networks we found that fibroblasts alter the characteristics of synchronous events. Although we cannot state how synchrony in basal ganglia areas *in vivo* would be affected by the presence of fibroblasts, we speculate that a similar change to that observed in the isolated neuronal network may shift the frequency distribution of pathologically active basal ganglia cells to a more physiological behavior in a similar way to the change caused by addition of dopamine or direct electrical stimulation. The precise way in which these antiparkinsonian treatments alter activity at the neuronal level is not understood. Direct electrical stimulation of cortical neuronal networks *in vitro* reduces rate of synchrony generation (Wagenaar et al., 2006), but dopaminergic stimulation *in vivo* changes the rate of synchronous bursting activity in STN to a higher frequency (> 75 Hz) than the beta frequency (11–30 Hz) associated with parkinsonian symptomatology (Brown et al., 2004; Brown et al., 2001).

In the unilateral 6-OHDA rat model, reduction in agonist-induced turning activity is a measure of normalization of basal ganglia function, and is observed following successful neuroregeneration (Dowd et al., 2005). Transplantation of fibroblasts to EP succeeded in reducing contralateral turning in response to apomorphine, but was not accompanied by dopaminergic regeneration. Meissner et al. (Meissner et al., 2000) did not find any affect of deep brain stimulation of the EP on striatal dopamine metabolism (levels of dopamine, 3, 4-dihydroxyphenylacetic acid and homovanillic acid) in naive and 6-OHDA-lesioned rats.

Administration of L-Dopa to hemiparkinsonian rats induces contralateral turning activity as well as dyskinesias which are very similar in appearance to those seen in human parkinsonian patients (Cenci et al., 2002; Lundblad et al., 2002), and the intensity of which increases with repeated drug administration (priming). Fibroblast transplantation to EP reduced dyskinesia severity in forelimbs and cranial area as well as reducing locomotor activity. Since increased synchronization of basal ganglia neuronal discharge is thought to be responsible for both L-Dopa-induced dyskinesia and akinesia (although this has not demonstrated conclusively), we propose that reduction of neuronal synchronization, or altered recruitment and distribution profile of synchronous events, is induced by the presence of fibroblasts in contact with the basal ganglia neurons, and that this altered synchronous activity is responsible for

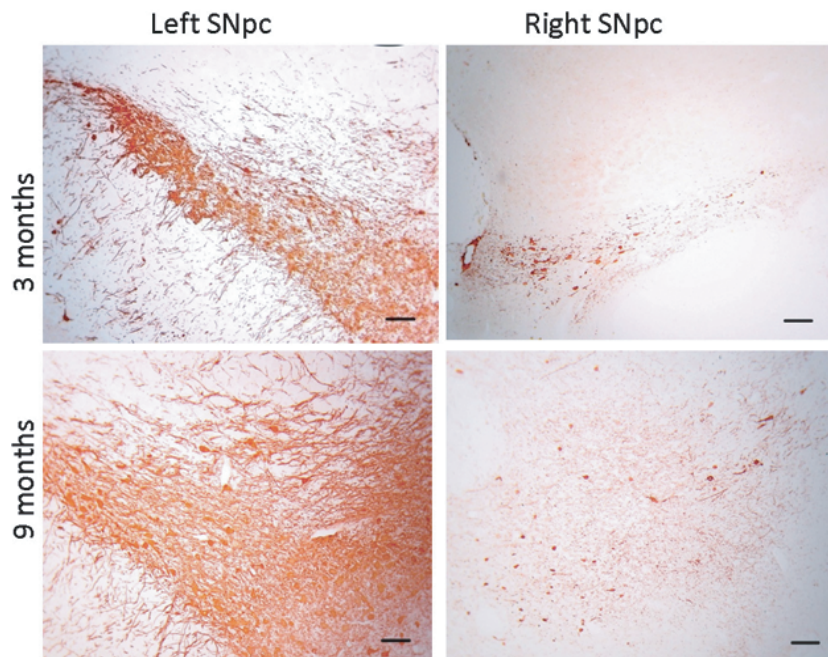


Fig. 7. Immunohistochemical staining for tyrosine hydroxylase in neurons of left and right substantia nigra, in rats administered 6-hydroxydopamine to right medial forebrain bundle. Animals were killed 3 or 9 months after implantation of syngeneic rat dermal fibroblasts to EP area. Tyrosine hydroxylase-positive neurons appear red. Horizontal bar represents 100 microns.

correction of the asymmetry defect in the hemiparkinsonian rat. By analogy, subthalamic DBS is associated with reduction of pathological beta activity in PD, and the suggested mechanism is modulation of pathological patterns of synchronized oscillations in the basal ganglia (Kühn et al., 2008; Bronte-Stewart et al., 2009).

Fibroblast transplantation could be normalizing parkinsonian symptomatology by a lesion-type effect, similar to that produced by pallidotomy, another procedure used occasionally in advanced PD, however a localized lesion of the EP by ibotenic acid did not affect DA agonist-induced turning or L-dopa-induced dyskinesia. Some researchers have found a decrease in apomorphine-induced circling by EP lesion in 6-OHDA-lesioned rats, (Dewar et al., 1983) but not all agree (Perier et al., 2003). In addition, fibroblast transplantation did not cause any general signs of malaise in the animals, whose body weight increased similarly to that of the controls, so that reduced turning and other movements were not the result of a general decrease in peripheral neuromuscular function.

Grafted fibroblasts were found on histological analysis 3, 7 and 9 months post transplantation. Staining for CD3, CD79a, and anti Iba-1 did not demonstrate active inflammation. We note that the fibroblasts caused a recruitment of astrocytes to the area of transplantation, as

normally seen in brain tissue damage. Both astrocytes and dermal fibroblasts express gap junction and multiple connexin channel types (Estacion, 1991; Salomon et al., 1994; Salomon et al., 1988) as well as potassium channels (Araque et al., 2001; Araque et al., 1999). Human dermal fibroblasts express BKCa channels that are known modulators of neuronal activity, and NO increases their opening through a sGC/cGMP/PKG pathway (Lim et al., 2005; Roh et al., 2007). The possibility exists, therefore, that BKCa channel opening could be activated in the transplanted fibroblasts in hyperactive basal ganglia by increased NO production, which could diffuse from nearby NOS-positive cells.

While the fibroblasts can produce a capacitance load and some active potassium current sink effect (Estacion, 1991; Salomon et al., 1988) the effects of astrocytes may be more complex. Astrocytes can release various neurotransmitters including glutamate (Parpura et al., 1994), D-serine (Wolosker et al., 1999) and ATP (Hassinger et al., 1996). Therefore the behavioral results observed could be the result of altered ion channel activity produced by both fibroblasts and astrocyte recruitment. The fact that microsphere transplantation had no effect on the parkinsonian rats indicates that physical obstruction to the course of an induced wave of depolarization (as occurs, for example, around

the bases of large blood vessels in cardiac atrial tissue during circus-like re-entry arrhythmia) is not associated with behavioral improvement. Thus we speculate that the improved behavioral effect of fibroblast implantation was the result of multiple factors including electrochemical coupling and structural changes in the transplanted area which probably resulted in alteration of the synchronized burst type activity seen in parkinsonian animal models in the STN-GPi-GPe pathway.

This study is limited by the fact that modification of the electrophysiological basal ganglia output by fibroblast transplantation was not demonstrated *in vivo*, which would have been further confirmation to our hypothesis and conclusions. These preclinical studies form the basis for further development of fibroblast transplantation technique as a clinical treatment. Although DBS is extremely effective in some patients, yet it is accompanied by occasionally severe side effects, including cognitive dysfunction (Temel et al., 2006) and is less effective in patients over 70 years old (Russmann et al., 2004), so that alternative treatments are required.

Transplantation of autologous fibroblasts to the human brain has already been carried out in attempts to treat Alzheimer's disease with cells modified to produce nerve growth factors, NGF and GDNF (Tuszynski, 2002; Tuszynski et al., 2005). Transplanted fibroblasts did not form tumors, or migrate to other parts of the brain, and no side effects were reported. The use of unmodified autologous fibroblasts transplanted to appropriate areas of human basal ganglia may therefore present a useful addition to existing therapies for PD, mainly in advanced stage disease accompanied by poor response to L-Dopa, even though it does not offer a cure for the disease. In addition, local neuronal electrophysiological modification by fibroblasts may be applicable for other clinical indications marked by neuronal hyperactivity.

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5.2. Disclosures

Dr. Finberg is a scientific advisor to GeneGrafts which is a Technion incubator company. Drs. Feld, Marom, and Gepstein are founders of GeneGrafts, Dr. Gluzman, Dr. Reshef, Dr. Loboda, Mr. Mohsen, Dr. Bressler-Stramer and Ms. Miari are former employees of GeneGrafts.

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